Markets

Lecture notes

What this lecture is about. Lecture 1 started from scarcity: not everyone can have everything. This lecture asks who gets a scarce good. The rule that decides is called an allocation mechanism, and a market is one such rule. We then define the benchmark market, perfect competition, by five conditions, and check four real markets against them: bitcoin mining, taxi licences, building AI models and selling AI answers. Along the way we meet barriers to entry. The last part introduces the two tools used for the rest of the course: demand and supply, each written as a function and as an inverse function.

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The lecture opened with an anonymous survey and a short guide to the slides (sign-in, chat and the pulse button). These notes skip them.

1 Review of lecture 1

Four short questions repeated the skills of lecture 1 and problem set 1 with new stories.

Review question 1: economic cost

You spend one summer making YouTube videos instead of doing an internship. You pay €200 to rent a camera and €60 for editing software. The internship would have paid €2,400. What is the economic cost of the YouTube summer? Options: €260, €2,140, €2,400, €2,660.

Step 1: explicit cost. This is what you pay: $200 + 60 = €260$.

Step 2: implicit cost. This is what you give up: the internship pay, €2,400.

Step 3: add them. Economic cost $= 260 + 2{,}400 = €2{,}660$.

€2,660. Economic cost includes both the explicit cost (€260) and the implicit cost (€2,400). The wrong options are typical mistakes: €260 counts only the money paid, €2,400 only the forgone pay, and €2,140 subtracts instead of adding.

Review question 2: the cost of tying up your money

A yearly gym pass costs €400, paid now. Or you can pay €430 at the *end* of the year. Your savings earn 5% a year. Which is cheaper, and by how much? Options: pay later; yearly, by €30; yearly, by €10; they cost the same.

Step 1: the saving. Paying now saves $430 - 400 = €30$.

Step 2: the lost interest. The €400 paid now cannot earn interest for the year. That costs $0.05 \times 400 = €20$.

Step 3: compare. The net saving is $30 - 20 = €10$.

Paying yearly (now) is cheaper, by €10. The answer "by €30" forgets the interest the money could have earned.

Review question 3: marginal benefit $\geq$ marginal cost

Each streaming subscription costs €10 a month. Assume you can put a euro value on each one: Netflix is worth €15 to you, Spotify €12, Disney+ €8 and Apple TV €5. Which do you keep? Options: all four; Netflix only; Netflix and Spotify; none.

Keep a subscription if its benefit is at least its cost of €10. Netflix ($15 \geq 10$) and Spotify ($12 \geq 10$) pass. Disney+ ($8 < 10$) and Apple TV ($5 < 10$) do not.

Netflix and Spotify. Disney+ and Apple TV are worth less than they cost.

Review question 4: marginal versus average

A YouTuber has made 10 videos so far, with 20,000 views on average. Video 11 gets 9,000 views. After video 11, does the average number of views fall, stay the same, or rise?

Step 1: total views. Ten videos with an average of 20,000 have $10 \times 20{,}000 = 200{,}000$ views. Video 11 adds 9,000, so the total is 209,000.

Step 2: new average.

$$\frac{10 \times 20{,}000 + 9{,}000}{11} = \frac{209{,}000}{11} = 19{,}000.$$

The average falls, to 19,000. The new video (the marginal one) is below the average, so it pulls the average down, but only a little: one video against ten. Two common mistakes: dividing 209,000 by 10 instead of 11 (20,900), and adding 20,000 and 9,000 instead of averaging (29,000).

2 Allocation mechanisms

A good is scarce when not everyone who wants it can have it. So someone or something has to decide who gets it.

Allocation mechanism

An allocation mechanism is the rule that decides who gets a scarce good.

The lecture used five rules:

1. **price**: whoever is willing and able to pay gets it;
2. **a test or contest**: whoever does best gets it;
3. **a government rule**: the state decides;
4. **the seller picks**: the seller chooses among the buyers;
5. **voting**.

Each of the following cases asks the same question, with these five rules as the options.

Case 1: ChatGPT Plus

The good is access to an AI model. Everyone who pays €20 a month gets it. What is the allocation mechanism?

Price. Whoever is willing and able to pay gets it.

Case 2: good grades

Teachers give out the good grades; every student wants them. Those who do best in the exam get them. What is the allocation mechanism?

A test or contest: the exam. A question to think about: what would go wrong if grades were sold for a price? They would stop telling anyone how able a student is.

Case 3: a job

A tech firm in Vienna offers an internship. It receives 200 applications for 2 places. The firm reads the CVs, conducts interviews and chooses. What is the allocation mechanism?

The seller picks. The employer chooses among the applicants (CV, interview, tests). The wage is a price too, but who gets the job is the employer's choice.

Case 4: a taxi licence in Istanbul

A taxi plate is the right to run one yellow taxi in Istanbul. The city fixes how many plates exist: about 17,400, the same number for decades. Existing plates are bought and sold for about €200,000 each. What decides how many taxis there are?

A government rule. The city decides how many plates exist; within that limit, a price decides who holds one. So two mechanisms work at once: a rule for the quantity and a market for the plates. Section 6 comes back to this market.

2.1 Markets

A market is an allocation mechanism too. Its rule: sellers offer a good at a price, and buyers decide whether to buy at that price. Examples are a farmers' market and a cryptocurrency exchange. To know who gets what in a market, we need the price and the quantities. Demand and supply (sections 9 to 11) give us exactly these.

3 Perfect competition

Perfect competition

A market is perfectly competitive when all five conditions hold:

1. **many small buyers and sellers** (a fragmented industry);
2. **price takers**: no one can move the price;
3. **identical products**: buyers don't care whom they buy from;
4. **perfect information**: everyone knows prices and quality;
5. **free entry**: anyone can start selling.

Perfect competition is the benchmark: real markets are compared with it. Note that "price taker" applies to buyers too, and to firms when they buy their inputs. In the next sections we check four markets against the five conditions. In each one, a condition can hold, hold only partly, or fail.

4 Bitcoin mining

Bitcoin mining

Computers compete to write the next page of bitcoin's public ledger, and get paid for it.

- **The ledger.** Every bitcoin payment is written into one public list. A new page (a *block*) is added about every 10 minutes.
- **The competition.** Special machines (ASIC chips) guess numbers billions of times a second. The first correct guess wins and writes the page.
- **The reward.** The winner gets 3.125 new bitcoin, plus the fees that senders attached to their payments.

4.1 The guessing race

A page contains the payments, the previous page's fingerprint and one number that the miner chooses. A fixed formula (SHA-256) scrambles the page into a code, the *hash*. The page only counts if the hash starts with many zeros. There is no shortcut: the miner changes the number, scrambles, checks, and repeats, again and again.

In class, students tried this on their phones. By hand it is hopeless. A phone finds a hash with 4 leading zeros in a few seconds: each digit of the hash has 16 possible values, so this takes about $16^4 = 65{,}536$, roughly 65,000, tries on average. Real bitcoin needs about 19 zeros in this notation, and all miners together make roughly 10^{21} guesses per second. So winning means making more guesses, and that means more machines and more electricity. Because each page contains the previous page's fingerprint, changing an old page would mean redoing all the work after it. This is the "chain" in blockchain.

Note that miners do not make the text "fit" the next block. They search for the number that makes the hash start with enough zeros.

Case 5: the mining reward

The good is 3.125 new bitcoin, every 10 minutes or so (from about 2028: 1.5625). It goes to the first miner whose guess makes the hash start with enough zeros. What is the allocation mechanism?

A contest: proof of work is the rule. Whoever first wins the guessing race gets the new coins. Your chance of winning equals your share of all guesses being made, so more machines (and more electricity) buy more chances. It works like a lottery in which every ticket is a guess, and every guess costs electricity.

Is bitcoin mining a perfect market?

Check the market for mining against the five conditions.

- Many miners: thousands, but most join one of a few pools.
- The price: 3.125 BTC per page, the same for every miner.
- Information: prices, difficulty and rewards are public.
- Starting: anyone can plug in a machine, if they have cheap power and thousands of euros for machines.

Which conditions fail? (Tick all that apply, or "none: it is perfect".)

condition		bitcoin mining has ...
many small sellers	partly	thousands of miners, but 3 pools together control more than half of all mining
price takers	holds	one world price for bitcoin; no miner can move it
identical product	holds	a valid page is a valid page; one bitcoin is like any other
perfect information	holds	price, difficulty and rewards are public, live
free entry	partly	anyone can plug in a machine, but cheap electricity and machines (€1,000–4,000 each) are not equally available to everyone

Two weak spots: many small sellers and free entry. Price takers, identical product and information are all fine. Verdict: close to perfect competition.

5 Barriers to entry

The weak spot "free entry" has a name.

Barrier to entry

A barrier to entry is anything that makes it hard for a new seller to enter a market.

Bitcoin mining shows four of them:

1. **Cheap electricity.** Miners pay about €0.05 per kWh in Texas or Paraguay; a household in Vienna pays about €0.25–0.30.
2. **Capital.** A professional bitcoin mine runs thousands of machines (ASIC chips) at about €1,000–4,000 each.
3. **Know-how and scale.** Cooling, buildings, contracts with power plants.
4. **Better machines.** Every new generation of machines makes the older ones unprofitable.

Monero: the laptop coin

Monero is a cryptocurrency designed so that normal computer chips (CPUs) can mine it, so entry is cheap. Your laptop makes a few thousand guesses per second, which earns roughly 10 cents of monero a day. A laptop running all day uses about 1 kWh, which costs about €0.25–0.30 in Vienna. Would mining monero on your laptop make you money?

Compare one day: you earn about €0.10 and pay about €0.25–0.30 for electricity. You lose roughly 15 to 20 cents a day.

No. The coins are worth less than the electricity (and the laptop gets hot). Entry is free, but the story is the same as for bitcoin: only miners with cheap power make money. (The numbers are rough. Special monero machines now exist too, so even here the advantage of ordinary chips is shrinking.)

6 Taxi licences

This is a market in which the number of sellers is fixed.

- **The market:** taxi plates. A plate is the right to run one yellow taxi in Istanbul.
- **Buyers:** people who want to run a taxi. **Sellers:** people who own a plate.
- **The number:** fixed at about 17,400 plates, the same for decades. The city issues no new ones.
- **The price:** about €200,000 per plate (11.25 million Turkish lira, October 2026).
- **For scale:** the minimum wage is about €510 a month, the average rent about €820, and a latte about €2.90. A plate costs about 33 years of minimum wage: 200,000/510 ≈ 392 months ≈ 33 years.

Two more facts. In 2024 the city announced 2,500 extra plates, and within days plate prices fell by about 6%. In New York, a taxi licence cost $200,000 in 2002 and $1.3 million in 2014; when ride apps came, its price fell by about 90%.

Why is a plate worth so much?

Options: the car is expensive; it lets you sell rides that others are not allowed to sell; the city pays the drivers; fuel is expensive.

It lets you sell rides that others are not allowed to sell. The plate's price is the value of being protected from competition. More plates, or ride apps, mean less protection, so the plate is worth less: this is what happened in Istanbul in 2024 and in New York.

Is the market for taxi plates a perfect market?

- Sellers: thousands of plate owners, but a fixed number, too few for a city this size.
- The price: about €200,000; buyers and sellers of plates take it as given.
- The product: a plate is a plate.
- Information: the going price is known (it is in the news).
- Starting: the city issues no new plates; you can only buy an existing one.

Which conditions fail?

condition		taxi plates have ...
many small sellers	partly	thousands of owners, but a fixed number: too few for a city this size
price takers	holds	about €200,000, taken as given
identical product	holds	a plate is a plate
perfect information	holds	the going price is in the news
free entry	fails	no new plates: only existing ones can be bought

Free entry fails, and many small sellers holds only partly. The only way in is to buy an existing plate for about €200,000. Price takers, identical product and information are fine. Verdict: not perfect competition.

6.1 What would make it a perfect market?

Free entry. Suppose the city gave a plate to anyone with a safe car and a clean record, with no cap on the number. Then there would be as many sellers as the city needs. What follows: more taxis, and a plate worth about €0, because anyone can get one.

This is a positive statement: it describes what happens to prices and quantities. Whether the city *should* do it is a normative question (lecture 1). Drivers and owners who borrowed money to buy a plate would lose, and safety and income are real concerns.

7 Building AI models

Who makes the AI models?

- **What is sold:** access to a model, for €20 a month, or at a price per million words (tokens).
- **How it is made:** training takes months on tens of thousands of chips. A top model costs hundreds of millions to billions of dollars.
- **Who sells:** a handful of labs. Examples are OpenAI, Anthropic, Google, xAI and Meta, and in China DeepSeek and Qwen.

Is the market for top AI models a perfect market?

- Sellers: a handful of labs.
- Prices: each lab sets its own (€20 and €200 tiers).
- The product: models differ in skill and style.
- Starting: billions of dollars, chips, experts and data.

Which conditions fail?

condition		top AI models have ...
many small sellers	fails	a handful of labs
price takers	fails	each lab sets its own prices (€20 and €200 tiers)
identical product	fails	models differ in skill and style
perfect information	partly	benchmarks exist, but quality is hard to judge
free entry	fails	billions of dollars, chips, experts and data

Almost all of them. There are few sellers, each sets its own price, the products differ, and entry costs billions. Information is only partly a problem, because benchmarks exist. Verdict: far from perfect competition, the opposite of bitcoin mining.

8 Selling inference

There is a second AI market: inference as a service.

- **Inference:** every time you send a prompt, a server computes the answer.
- **Open models:** models that anyone can download (DeepSeek, GLM, Kimi).
- **The sellers:** a dozen or so firms run the *same* open model and sell inference (for example Together AI and Cloudflare; OpenRouter lists them side by side), plus the lab's own service. According to one analysis, 16 sellers offer DeepSeek on OpenRouter.
- **You buy** from whoever is cheapest or fastest.

What do the sellers compete on? How the chips (GPUs) are set up to run a model decides how fast and how cheap each answer is, and the hosts compete on this. The lab knows its own model best, so it could run it fastest and cheapest. But the lab puts most of its GPUs into training the next model: its own service is cheap, but not the fastest, and sometimes full. The hosts bring more servers and their own know-how to make the model fast and cheap, and they offer many models under one roof.

Why buy from a host?

The lab and a host sell the same model at the same price. Why would anyone buy from the host? Options: the host gives a better model; the host answers faster and has room for you; the law requires it; no reason.

The host answers faster and has room for you. With the same model and the same price, what differs is speed and capacity. The lab's GPUs go to training, so the hosts fill the gap with servers and know-how, and with many models in one place. That is what they compete on, and it is why the product is not quite identical.

Is inference as a service a perfect market?

- Sellers: a dozen or so hosts per open model, plus the lab itself.
- The product: the same model, but some hosts answer faster and/or cheaper.
- Prices: they vary between providers; all information is public.
- Starting: anyone may start, but GPUs are expensive, and so is the electricity.

Which conditions fail?

condition		selling inference has ...
many small sellers	fails	a dozen or so hosts per model: few, not many
price takers	holds	charge more than the others and customers leave
identical product	partly	the same model, but the speed differs
perfect information	holds	prices are listed side by side
free entry	partly	anyone may start, but GPUs are expensive to rent, or to buy and run (electricity again)

Many small sellers fails (a dozen or so is few), and identical product and free entry hold only partly. Price takers and information are fine. Verdict: still close to perfect competition. Compare with section 7: building a top model is far from perfect, while selling answers from an open model is close to it.

9 Demand and inverse demand

From here on we treat all consumers of a good as one group.

Demand

Demand: what quantity do consumers want to buy at a given price?

As a function we write $Q^D = f(P)$: pick a price, get the quantity demanded. When the price falls, more people buy.

Inverse demand

Inverse demand: what price are consumers willing to pay for a given quantity?

9.1 A demand function

Take an AI subscription, with the price P in euro per month and the quantity Q in thousands of subscribers. The demand function is

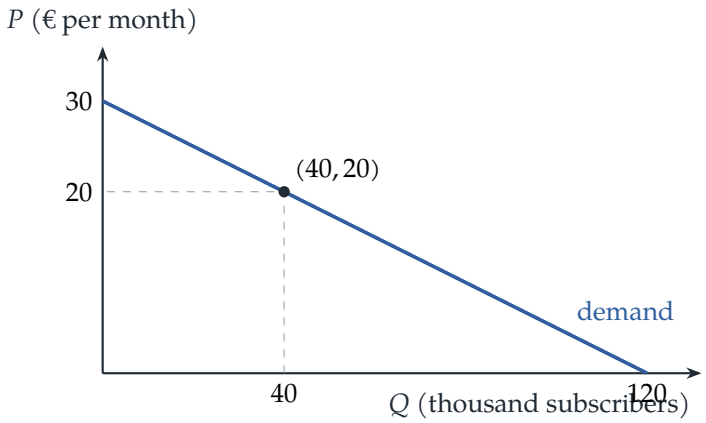
$$Q = 120 - 4P.$$

- **120:** if it were free ($P = 0$), 120,000 people would subscribe.
- $-4P$**:** each €1 more a month means 4,000 fewer subscribers.
- **At €20:** $Q = 120 - 4 \cdot 20 = 40$, so 40,000 subscribers.

To get the inverse demand, solve for P:

$$Q = 120 - 4P \implies 4P = 120 - Q \implies P = 30 - \frac{Q}{4}.$$

- **30:** at €30 or more, nobody subscribes. Check: $Q = 0$ gives $P = 30$. This is the *choke price*.
- $-Q/4$**:** to win 4,000 more subscribers, the price must fall by €1.



The demand curve $Q = 120 - 4P$, drawn with P on the vertical axis. At €20 a month, 40,000 people subscribe.

As is usual in economics, the price is on the vertical axis. Read the picture from the price axis across to the curve, and it shows demand. Read it from the quantity axis up to the curve, and it shows inverse demand.

Practice: inverse demand

Printing at the campus copy shop, with P in cents per page and Q in pages per day. The demand is $Q = 160 - 4P$: if printing were free, students would print 160 pages a day, and each extra cent per page means 4 fewer pages a day. What is the inverse demand? Options: $P = 160 - 4Q$; $P = Q/4 - 40$; $P = 40 - Q$; $P = 40 - Q/4$.

Move the P-term to one side, then divide *every* term by 4:

$$Q = 160 - 4P \implies 4P = 160 - Q \implies P = 40 - \frac{Q}{4}.$$

Check: $Q = 0$ gives $P = 40$ in the inverse form, and $P = 40$ gives $Q = 160 - 160 = 0$ in the original form.

$P = 40 - Q/4$. Read it: at 40 cents a page or more, nobody prints; to sell 4 more pages a day, the price must fall by 1 cent. The other options are typical mistakes: swapping P and Q, a sign error, and dividing only the constant by 4.

10 The law of demand

The law of demand

When a good's price rises, people buy less of it, all else equal.

- **Demand form:** price up ⇒ quantity demanded down (all else equal).
- **Inverse form:** the marginal willingness to pay decreases as the quantity increases.
- **Why:** each extra unit is worth less to you than the one before.
- **The picture:** a downward-sloping curve, as in the figure above.

"All else equal" matters. When something other than the price changes, the whole curve moves. That is the topic of the next lecture.

11 Supply and inverse supply

Supply

Supply: what quantity do sellers want to sell at a given price? It slopes up because each extra unit costs more to provide.

- **Any firm:** producing more needs overtime, extra staff and new machines, so each extra unit costs more (rising marginal cost).
- **Bitcoin miners:** when the bitcoin price is low, the reward barely pays for the electricity, so fewer miners mine.

As a function we write $Q^S = f(P)$: pick a price, get the quantity supplied.

Inverse supply

Inverse supply: what price do sellers need to offer a given quantity?

11.1 A supply function

Take the market for used mining machines, with P in euro per machine and Q in machines offered per month. The supply function is

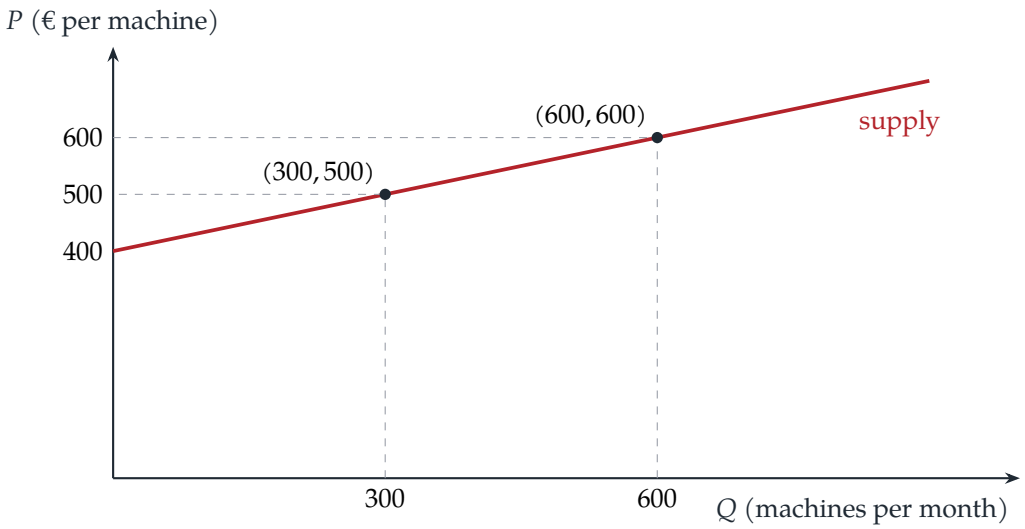
$$Q = 3P - 1200 \qquad (\text{for } P \geq €400).$$

- $3P$**:** each €1 more means 3 more machines offered a month.
- -1200**:** this has no meaning on its own. It only puts the start of the curve at €400, since $1{,}200 = 3 \times 400$.
- **At €500:** $Q = 3 \cdot 500 - 1200 = 300$ machines.

Solving for P gives the inverse supply:

$$Q = 3P - 1200 \implies 3P = Q + 1200 \implies P = 400 + \frac{Q}{3}.$$

- **400:** at a price below €400, nobody sells ($Q = 0$). Owners would rather keep mining with their machines.
- $+Q/3$**:** the more machines you want offered, the higher the price must be. 300 machines need $400 + 300/3 = €500$; 600 machines need $400 + 600/3 = €600$.



The supply curve $Q = 3P - 1200$, drawn as $P = 400 + Q/3$. It starts at €400 and slopes up.

The same market for used mining machines comes back in the next lecture, together with its demand.

Practice: inverse supply

Dog-walking by students, with P in euro per walk and Q in walks per week. The supply is $Q = 5P - 15$. What is the inverse supply? Options: $P = 3 + Q/5$; $P = 3 + Q$; $P = Q/5 - 3$; $P = 5Q - 15$.

Bring the 15 to the other side (it becomes +15), then divide every term by 5:

$$Q = 5P - 15 \implies 5P = Q + 15 \implies P = 3 + \frac{Q}{5}.$$

Check: at $P = 3$, $Q = 5 \cdot 3 - 15 = 0$: nobody offers a walk.

$P = 3 + Q/5$. Below €3 a walk, no student offers to walk a dog; each extra walk a week needs the price to rise by €0.20.

12 Summary

- **Allocation mechanism:** the rule for who gets a scarce good: price, a test or contest, a government rule, the seller picks, or voting. Markets use prices.
- **Perfect competition:** many small sellers, price takers, identical products, perfect information and free entry, all at once.
- **Barriers to entry:** anything that keeps new sellers out, such as cheap power available only to some, or the capital for machines.
- **Four markets:** bitcoin mining and selling inference are close to perfect competition; building top AI models is far from it; taxi plates are not perfectly competitive, because entry is closed. A plate's price is the value of protection from competition.

- **Demand** $Q^D = f(P)$: what quantity consumers want to buy at a given price. It slopes down (the law of demand).
- **Inverse demand:** what price consumers are willing to pay for a given quantity. Example: $Q = 120 - 4P$ gives $P = 30 - Q/4$.
- **Supply** $Q^S = f(P)$: what quantity sellers want to sell at a given price. It slopes up, because each extra unit costs more.
- **Inverse supply:** what price sellers need to offer a given quantity. Example: $Q = 3P - 1200$ gives $P = 400 + Q/3$.
- **To invert:** move the P-term to one side, then divide every term by its coefficient. Check with $Q = 0$.